



Open Inc.

Data Science Training Program



DS Training Program

4-Month (16-Week) Data Science Training Syllabus

 Duration: 4 Months (16 Weeks)

 Class Timing: 2 Hours per Day

• Syllabus Overview

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Program Overview

(At a Glance)



DURATION

4 Months (16 Weeks)



CLASS TIMING

2 Hours per Day



ASSESSMENTS

Every Friday (Weekly Assessment)

★ OUTCOME

Strong Data Science foundation with hands-on ML project

Tools & Tech

Python

Statistics

Pandas

NumPy

Matplotlib

Seaborn

SQL

Machine Learning (Basics)

Power BI / Tableau (Intro)



Month 1 - Python & Statistics Fundamentals



Week 1: Introduction to Data Science & Python Basics

- What is Data Science?
- Data Science vs Data Analytics
- DS lifecycle
- Python installation & setup
- Python basics: variables, data types
- Operators
- Input/output

FRIDAY ASSESSMENT

Python basics MCQs + coding

Month 1 — Weeks 2–4 Details



Week 2: Python Control Structures & Functions

- Conditional statements (if, elif, else)
- Loops (for, while)
- Functions
- Lambda functions
- Exception handling
- File handling basics

FRIDAY ASSESSMENT

Python logic building test

Week 3: Python Data Structures

- Lists, Tuples, Sets, Dictionaries
- List comprehensions
- Dictionary operations
- Working with strings

FRIDAY ASSESSMENT

Data structures coding test

Week 4: Statistics for Data Science

- Types of data
- Measures of central tendency
- Measures of dispersion
- Probability basics
- Distributions (Normal, Binomial, Poisson)
- Correlation & covariance

FRIDAY ASSESSMENT

Statistics fundamentals test

Month 2 - Data Analysis with Python & SQL



Week 5: NumPy & Pandas Basics

- NumPy arrays
- Array operations
- Pandas Series & DataFrames
- Reading CSV, Excel files
- Data selection & indexing



FRIDAY ASSESSMENT
NumPy & Pandas practical

Week 6: Data Cleaning & Preprocessing

- Handling missing values
- Handling duplicates
- Data type conversion
- Outlier detection
- Feature scaling (basic)



FRIDAY ASSESSMENT
Data cleaning case study

Month 2 — Visualization & SQL



Week 7: Data Visualization

- Matplotlib basics
- Seaborn basics
- Line, Bar, Histogram, Box plot
- Heatmaps
- Data storytelling



FRIDAY ASSESSMENT

Visualization assignment



Week 8: SQL for Data Science

- SQL basics
- SELECT, WHERE, ORDER BY
- Aggregations
- GROUP BY, HAVING
- Joins



FRIDAY ASSESSMENT

SQL practical test

Month 3 - Machine Learning Fundamentals

Week 9: Introduction to Machine Learning

- What is Machine Learning?
- Types: Supervised, Unsupervised
- ML workflow
- Train-test split
- Model evaluation basics

📌 Friday Assessment: ML concepts test

Week 10: Supervised Learning Algorithms

- Linear Regression
- Multiple Linear Regression
- Logistic Regression
- KNN
- Model evaluation metrics

📌 Friday Assessment: Regression & classification task

Month 3 - Unsupervised & Model Improvement

Week 11

Unsupervised Learning

- Clustering concepts
- K-Means clustering
- Hierarchical clustering
- Dimensionality reduction (PCA – intro)

 Friday Assessment: Clustering assignment

Week 12

Model Improvement & Intro to Advanced Topics

- Feature engineering
- Overfitting & underfitting
- Cross-validation
- Hyperparameter tuning (intro)
- Intro to Time Series / NLP (overview)

 Friday Assessment: ML case study

Month 4 - Project Planning & Implementation

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Week 13: Project Planning

- Business problem understanding
- Dataset selection
- EDA planning
- Tools selection



Week 14: Project Implementation

- Data cleaning
- EDA using Python
- Model building

Month 4 - Finalization & Career Readiness



Model Evaluation

WEEK 15

- **Model Tuning**

Fine-tuning hyperparameters to optimize performance metrics.

- **Visualization of Results**

Creating compelling visual representations of model outcomes.

- **Insights Generation**

Deriving actionable business insights from data analysis.



Career Readiness

WEEK 16

- **Project Presentation**

Showcasing the capstone project findings and methodology.

- **Resume Building**

Tailoring resumes specifically for Data Science roles.

- **Interview Prep**

Technical questions on Python, Stats, ML, and mock interviews.

Assessments & Evaluations



WEEKLY SCHEDULE

Weekly Assessments: Every Friday



FINAL MILESTONE

Final Evaluation: Capstone Project + Viva

Weekly Assessment Examples

- ✓ Python basics MCQs + coding
- ✓ Python logic building test
- ✓ Data structures coding test
- ✓ Statistics fundamentals test
- ✓ NumPy & Pandas practical
- ✓ Data cleaning case study
- ✓ Visualization assignment
- ✓ SQL practical test
- ✓ ML concepts test
- ✓ Regression & classification task
- ✓ Clustering assignment
- ✓ ML case study

Student Deliverables



Weekly assessments



Python notebooks



Visualization reports



One ML-based capstone project



GitHub project repository (optional)

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**Deliverables across learning
+ project outputs**

